Course milestone M4: Research Contract v0 and Permission Determination

HONR 46400 · Evidence-Driven Research

What to submit

One file, `lastname_m04.pdf`. Everything this milestone asks for goes inside it, as sections in the order below.

Submit it on Brightspace, under **Assignments** then **M04**. Brightspace carries the deadline.

Your work lives in a Colab notebook, which has no export-to-PDF button. **Making the PDF you hand in**, at the end of this document, is the one-minute routine that turns the notebook into the file you upload.

This milestone presents **Book Milestone 4: Your research contract, v0** (checkpoint, version 1) of the course book, EDR|AI.

What this course adds

Your first mentor meeting, requested

Every project in this course has a faculty mentor of record, and that mentor is your instructor. The Undergraduate Research Expo application you file later names that mentor, and a mentor who has never discussed the project cannot stand behind it. So the course requires **two mentor meetings**, one in each half of the semester, and this milestone opens the first of them.

Request the Round 01 meeting. Email your instructor from your Purdue address, asking for a meeting inside the Round 01 window your course platform posts. The email carries three things:

1. **At least three concrete slots**, each a specific weekday and starting hour inside that window. Offering options is the request; asking when the instructor is free is not.
2. **Your project in two sentences**: the question you have declared, and where the work currently stands.
3. **The one decision you most want pressure-tested**, named outright. A meeting with a stated agenda changes a project. A meeting without one only reports on it.

Save the sent email as a PDF and carry it as a section of your milestone document.

What Round 01 is for. It happens early on purpose, while changing direction is still cheap. It works on your question, the evidence behind it, and the contract you have just declared. It is the meeting where a project that cannot be carried gets replaced before an analysis is built on top of it.

Requesting the meeting is what this milestone collects. Holding it, and reporting that it happened, belong to a later milestone. If your project is an approved group, at least one member attends each meeting, and it need not be the same member at both rounds.

Milestone 4: Your research contract, v0

Open the milestone workbook in Colab

This chapter closes Studio 4: Declare and diagnose provisionally. Its lessons each ended at **It is your turn**; what you wrote there arrives here as working material, and this is where the pieces become one artifact you can defend.

What this milestone produces. Research Contract v0: objective, target estimand, population, setting and time, data strategy, a provisional operationalization you mark for revision, warrant, answer strategy and uncertainty statement, plus the design properties you diagnosed (its bias, its wobble, and how often it would detect what you are looking for), your permission status, and a redesign record. Measurement proper is taught and assessed in Studio 6; here you only commit to a starting choice and say why it is defensible for now.

What you bring

You also carry forward the last milestone's artifact: **Milestone 3: Your evidence base**. This one builds on it, and the chain assumes it is versioned and filed.

Check you are carrying each of these before you start. If one is missing, go back and work that lesson's **It is your turn** first; the practice below assumes it.

- Lesson 10: Model, Inquiry, Data Strategy, and Answer Strategy: your MIDA declaration: model, inquiry, data strategy, and answer strategy, with the alignment sentence that ties them , carrying population, setting and time, the provisional operationalization, and the warrant into Contract v0.
- Lesson 11: Uncertainty Before You Need It: your estimand and estimator, the repeat-run story behind them, your dependence structure, and your uncertainty sentence next to its wrong twin.
- Lesson 12: Declaring and Diagnosing a Research Design: your design diagnosis: bias, wobble, and power read from simulation, one redesign, and the honest call that followed.
- Lesson 13: Research Ethics and Data Governance: your declared permission status, your minimised column list, your AI boundary, and your four governance decisions.

The practice

1. Write the four MIDA parts of your design so a stranger could run them. (MIDA is Blair, Cooper, Coppock, and Humphreys' framework, developed at book length in RDSS.)
2. State your estimand and estimator, and describe in words what would differ if the procedure ran again.
3. Diagnose the design: what is its bias, how much does it wobble, and how often would it detect what you are looking for.
4. Run the permission determination and declare one status; if it is anything but cleared, name the authority and the date you asked.
5. Minimise your columns against the declared analysis, and check what a re-identification test would find.
6. Write the redesign record - what you changed after diagnosing, and why. A diagnosis that changed nothing is a diagnosis you should distrust.

A version, not a pass

Your milestone artifact is a dated, numbered **version** with the reason for the version attached. When later evidence changes it, you write the next version rather than editing the last one, because the sequence of changes is itself part of your research record.

The four rails, here

Four concerns cross every studio in this book. At this milestone they take these specific forms.

- **Ethics, permissions, and data exposure.** The permission status produced here gates every later studio; nothing downstream may proceed past a stop.
- **Evidence, provenance, and reproducibility.** Your data strategy names actual sources from the Studio 3 registry, not source types.
- **AI activity, verification, and human decisions.** Diagnosis is delegable; the choice of what to fix is not.
- **Uncertainty, claim boundary, and revision history.** This is where your uncertainty statement is first written down and first tested.

Where this milestone sits

You arrived carrying Milestone 3: Your evidence base. Milestone 5 commits the contract to one research pathway and writes what that pathway can and cannot establish. The chain ends at Milestone 12: Your release and next cycle, where you decide whether your finished research artifact leaves your hands.

What sends you back

Route diagnosis, measurement changes, permission rulings, and the realized data each produce a new Contract version with its reason.

How this milestone is assessed

Each row scores **0**, **1**, or **2**. **10 points total.**

#	Criterion	0	1	2
1	The milestone artifact exists in full, specific to your own project	absent, or generic text that would fit any project	present but incomplete, or specific in some parts and generic in others	complete and unmistakably about your project
2	The milestone is written as a dated, numbered version with its reason	no version, or a version with no reason given	versioned, but the reason is decorative rather than an account of what changed	versioned with a reason a reader could use to reconstruct your thinking
3	Every judgment in the artifact is defended by you, not asserted by a tool	conclusions appear with no reasoning, or reproduce a tool's wording	reasoning present but thin where the decision was hardest	each judgment carries a reason you could defend under questioning
4	The four rails are carried in the form this studio requires	one or more rails absent where this studio required them	all rails present but one is nominal	every rail addressed in the form this studio requires
5	The milestone's own product is present and defensible: Research Contract v0: objective, target estimand, population, setting and time, data strategy, a provisional operationalization marked for revision, warrant, answer strategy and uncertainty statement, plus the design properties you diagnosed (its bias, its wobble, and how often it would detect what you are looking for), your permission status, and a redesign record	the milestone's own product is missing	present but would not survive a careful question	present and defensible as stated

> ! Blocking gate
>
> No work at or after Studio 4 proceeds past a not-authorized determination. This is not scored and cannot be averaged away.

Your workbook

Open the workbook with the link above. It walks these steps with a cell for each one, ends by writing your milestone version with its reason, and adds the rows this studio contributes to your AI Research Ledger and your Research Dossier.

Making the PDF you hand in

Colab has no export-to-PDF button. You make the PDF by **printing the notebook to a file**, which takes about a minute once you know the preparation steps. Skipping them is the most common way a milestone arrives looking half finished, because a notebook prints what is on the screen and nothing else.

1. Prepare the notebook

1. **Run the whole thing.** `Runtime` ⟶ `Run all`, then wait for it to finish. A cell you never ran prints with nothing underneath it, so an analysis you really did can reach the grader as an empty box.
2. **Expand every collapsed section.** Anything folded shut prints as a heading with nothing beneath it. Open all of them, including any Colab folded for you.
3. **Shorten any output that scrolls.** When Colab puts a scrollbar on an output, or tells you the output is truncated, only the visible part prints. Show the first twenty rows instead of the whole table, or summarise it, then rerun that cell. A reader did not want the whole table anyway.
4. **Read it once, top to bottom.** You are about to freeze it.

2. Print it to a file

`File` ⟶ `Print` opens your browser's print dialog. Set the destination to **Save as PDF**, open the dialog's further settings, and turn on **Background graphics**, so code cells keep their shading and your figures keep their backgrounds. Save it under the file name this milestone asks for.

Wide tables and wide figures are the usual casualty. If something runs off the right edge, switch the layout to **landscape** or reduce the scale, and print again.

3. Check the PDF before you upload it

Open the file you just made and confirm three things: every figure is whole rather than sliced by a page break, no output stops mid-line, and the pages run in order with nothing missing. A PDF is an artifact like any other in this course, so you verify it before your name goes on it.

One route not to take

Ask an AI assistant how to export a Colab notebook and it will very likely hand you `jupyter nbconvert --to pdf`. Do not spend your afternoon on it. Inside Colab that route needs a large LaTeX installation, runs for several minutes, and then either fails or quietly drops the symbols and emoji these notebooks are full of. Printing to a file is the route that works.

That is a small, cheap instance of the rule this whole course runs on. The confident answer is not automatically the correct one, and the way you find out is to check it against what actually happens.